

Practice Quiz 05D — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 05

Choosing and proving a contrapositive.

1. Prove by contraposition that, for all real numbers a, b ,

$$a + b > 10 \longrightarrow (a > 5 \vee b > 5).$$

State the contrapositive explicitly before proving it.

Solution Let a, b be arbitrary real numbers. De Morgan's law shows that the contrapositive is

$$(a \leq 5 \wedge b \leq 5) \longrightarrow a + b \leq 10.$$

Suppose $a \leq 5$ and $b \leq 5$. Adding the two inequalities gives $a + b \leq 5 + 5 = 10$. This proves the contrapositive, so the original implication follows. Since a, b were arbitrary, it holds for all real a, b . ■